

YASH ARYAN

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PROFESSIONAL SUMMARY

Computer Science undergraduate pursuing a B.Tech in Cloud Computing and Automation (CGPA: 8.0/10), with a strong foundation in C++, Data Structures Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Linux/Unix, and distributed systems. Experienced in developing real-time, scalable software using C++, Go, Node.js, WebSockets, Redis, PostgreSQL, Docker, and AWS. Solved 500+ DSA problems and built production-grade systems involving concurrency, networking, authentication, distributed messaging, and system debugging. Interested in software engineering for embedded platforms, Linux-based systems, connectivity, and performance-critical applications.

TECHNICAL SKILLS

Programming Languages: C++, C, Java, Python, Go, JavaScript, TypeScript, SQL

Core Computer Science: Data Structures Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Concurrency, Multithreading, System Design, Debugging

Systems Linux: Linux/Unix, Shell Scripting, Unix Tools, Process Management, Networking Fundamentals, Git

Networking Distributed Systems: TCP/IP, HTTP/REST, WebSockets, Redis Pub/Sub, Client-Server Architecture, Distributed Messaging

Backend Systems Development: Node.js, Express.js, FastAPI, REST APIs

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB

Cloud DevOps: AWS (EC2, S3, DynamoDB), Docker, Docker Compose, CI/CD

Frontend: React.js, Vite, TypeScript, Tailwind CSS, PWA

EDUCATION

Vellore Institute of Technology (VIT Bhopal)

Expected Graduation: Sept 2028

B.Tech in Computer Science, Specialization in Cloud Computing and Automation **CGPA: 8.0 / 10.0** Relevant Coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Object-Oriented Programming, Cloud Computing Tools & Services, AI Agents, Unix Tools and Scripting

PROJECTS

VIT Live — Real-Time Campus Engagement Platform

Personal Project | GitHub

Go (Fiber), React 19, TypeScript, PostgreSQL, Redis Pub/Sub, WebSockets, Razorpay, Tailwind CSS, Framer Motion, PWA

- Architected a full-stack real-time platform with 5-role RBAC, live announcements, emergency alerts, polls, and club feeds using WebSockets and Redis Pub/Sub, achieving sub-second message delivery.
- Integrated Razorpay payments with HMAC-SHA256 signature verification, QR-code event ticketing, and live attendee check-in.
- Implemented JWT authentication with rotating refresh tokens, OTP verification, anonymous polling, audit logging, and CI-tested REST APIs.
- Developed an installable React PWA with offline support, Firebase Cloud Messaging push notifications, admin dashboard, scheduled publishing, and S3-compatible object storage across a 13K+ LOC monorepo.

Unified Health Intelligence Platform (UHIP) — Cloud-Native Healthcare Platform

Team Project

- Designed and deployed a multi-service software architecture integrating React, Express.js, FastAPI, PostgreSQL, and MongoDB services using Docker and Docker Compose.
- Configured container networking, persistent storage, service dependencies, environment management, and inter-service communication for a reproducible Linux-based deployment environment.
- Performed system-level debugging, integration testing, failure analysis, and deployment troubleshooting to improve reliability across multiple services.
- Led containerization and deployment activities for the team project; awarded an S Grade and presented the system at the College Project Exhibition.

ColCode — Real-Time Collaborative Code Editor

Personal Project | Production Docker Deployment

React, TypeScript, Monaco Editor, Yjs (CRDT), WebSockets, xterm.js, Node.js, Fastify, PostgreSQL, Redis, Docker, Caddy, Puppeteer

- Built a real-time collaborative code editor using Yjs CRDT and WebSockets, enabling conflict-free concurrent editing with live cursors, selections, user presence, and zero lost keystrokes during simultaneous typing.
- Engineered a secure code-execution sandbox where each run launches a single-use Docker container with network isolation, read-only root filesystem, dropped capabilities, and cgroup CPU, memory, PID, and wall-clock limits; supports Python, JavaScript, TypeScript, Ruby, and Go with streamed stdout/stderr.
- Implemented GitHub OAuth, JWT sessions, and server-side project RBAC (owner/editor/viewer) with authorization enforced during WebSocket upgrades; persisted CRDT snapshots to PostgreSQL to maintain project state across server.
- Load-tested the synchronization server at 3,500 ops/s across 100 rooms and 500 concurrent clients, achieving 100 shipped 150 automated checks covering synchronization, access control, sandbox limits, browser E2E testing, and production Docker Compose deployment behind Caddy with HTTPS.

CERTIFICATIONS & ACHIEVEMENTS

- **Applied Machine Learning in Python** — University of Michigan (Coursera): Predictive Modeling, Scikit-Learn.
- **Cloud Computing and Distributed Systems** — NPTEL, IIT Kanpur (Elite + Silver, 78%): Virtualization, Scalability.
- **Introduction to Internet of Things** — NPTEL, IIT Kharagpur (Elite, 77%): IoT Architecture, Embedded Systems, Protocols.
- **Solved 500+ Data Structures & Algorithms problems** on LeetCode and GeeksforGeeks; following Striver's A2Z DSA Sheet.